LESSON 4.3

Estimate the Percent of a Number

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can estimate the percent of a number using benchmark percents, rounding, and compatible numbers.

WHY THIS MATTERS Margaret checked the cloud storage on her phone: 6 GB total, 4 GB used. Approximately what percent of her storage is used — and of the used storage, approximately what percent are photos?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
estimate *Spanish: estimación*	A reasoned answer that is close enough to be useful, found without exact computation.	about 70% used
benchmark percent *Spanish: porcentaje de referencia*	A friendly percent — 1%, 10%, 25%, 50%, 75%, 100% — used as a landmark for estimating.	10%, 25%, 50%, 75%
compatible numbers *Spanish: números compatibles*	Numbers close to the originals that are easy to compute with mentally.	8.6% of 216 → 10% of 220: both nudged to numbers that divide nicely.
double number line *Spanish: recta numérica doble*	Two parallel number lines that pair amounts with their percents at the same positions.	0, 30, 60 GB over 0%, 50%
the whole *Spanish: el todo*	The quantity that counts as 100% in a problem — and it can change between questions.	all of the storage
Decompose *Spanish: Descomponer*	Break a percent into friendly pieces that are easy to find.	15% = 10% + 5%

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A game download is 80 MB. 41 MB are done. About what percent is done?

1. Whole: 80 MB is 100%.
2. 50% of 80 is 40 MB.
3. 41 MB is just over 40 MB.

Answer: about 50%

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

The Green-Team recycled 216 pounds of material; 8.6% was metal.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The whole in this problem is _____ , so it stands for one hundred percent.
- The benchmark closest to _____ is _____ .
- My estimate is about _____ % because _____ .
- I know my estimate is reasonable because _____ .

Word bank estimate | about | benchmark | whole | part | double number line | half | one quarter | three quarters | close to | reasonable | compare

WATCH OUT!

A common mistake

Keeping the same whole for every question

LESSON 4.3

Estimate the Percent of a Number

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

Margaret's phone has 6 GB of storage and 4 GB is used. About what percent is used?

(A) About 25%
(B) About 40%
(C) About 70%
(D) About 150%

STEPS TO TRY
1. What is the whole?
2. 4 out of 6 is the same as 2 out of 3.
3. Two-thirds is close to which benchmark?

SHOW YOUR WORK

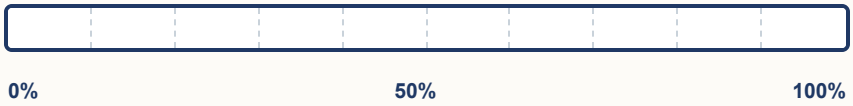
2 **Circle the letter of the best answer. Show how you know.**

On the double number line for 6 GB, what amount lines up with 25%?

(A) 1.25 GB
(B) 1.5 GB
(C) 2.5 GB
(D) 4 GB

STEPS TO TRY

1. 25% means one quarter.
2. One quarter of 6.
3. 6 ÷ 4.



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

3 **Circle the letter of the best answer. Show how you know.**

Which benchmark percents make good landmarks on a number line?

(A) 0%, 25%, 50%, 75%, 100%
(B) 3%, 17%, 62%, 94%
(C) Only 50%
(D) Only 100%

STEPS TO TRY

1. Which percents can you find mentally?
2. Quarters and tenths are the friendly ones.
3. 0, 25, 50, 75, 100.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

To estimate 8.6% of 216 by rounding, which numbers should you use?

(A) 1% of 216

(B) 8% of 300

(C) 10% of 220

(D) 86% of 216

STEPS TO TRY

1. Round each number to something friendly.
2. 8.6% is close to which benchmark?
3. 216 is close to which round number?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

You sorted a move into 'Benchmark percent' and your partner into 'Compatible numbers.' Which number did each strategy change — the percent or the whole — and why does changing that one keep the estimate close?

SENTENCE STARTER This is ___ because it changes the ___, and the estimate stays close since ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help